

QSVM Track Progress Summary

1 Overview

This report summarizes progress on the QSVM (quantum support vector machine) malware classification track, covering EMBER 2018 sweeps, a subsample-sizing decision, an angle-vs-IQP encoding comparison, and a SOREL-20M labelling decision. Both binary and multiclass classification results are included below.

2 Binary Classification Results

2.1 EMBER binary classification

On EMBER, QSVM performance sits near chance while classical SVM performs substantially better and continues to improve with more components.

$n_{\text{components}}$	QSVM (angle)	QSVM (IQP)	SVM
1	0.4532	0.4587	0.5314
3	0.4924	0.5084	0.6306
6	0.5138	0.5137	0.6724

Table 1: EMBER binary classification accuracy, QSVM (angle/IQP encoding) vs. classical SVM.

2.2 Significance

An earlier conclusion — that QSVM ties classical SVM on binary classification — was based solely on CIC-MalMem, where QSVM reached ~ 0.99 . The EMBER results show that this apparent tie was a ceiling effect specific to an easy, highly separable dataset, not a general property of the method: on a second, harder dataset the gap between QSVM and classical SVM is large and does not close as more components are added.

3 Multiclass Classification Results

3.1 EMBER 15-class

Held-out macro-F1 (mean over repeated outer folds) for the 15-class EMBER family task:

QSVM correctly recognizes 9 of 15 families at $n_{\text{components}} = 1$, and 14–15 of 15 families from $n_{\text{components}} = 3$ upward, with no collapse toward random guessing. This is notable because EMBER’s 15 families are exactly balanced (955 rows each), whereas the CIC-MalMem 15-class task is only mildly imbalanced ($\sim 1.7\times$) yet QSVM collapses to near-random performance there. Since

$n_{\text{components}}$	QSVM (angle)	QSVM (IQP)	SVM
1	0.1688	0.1541	—
3	0.3908	0.4797	—
6	0.4232	0.5195	0.7930

Table 2: EMBER 15-class held-out macro-F1. SVM baseline only available at $n_{\text{components}} = 6$.

EMBER is comparably or less balanced in relative terms and does not collapse, class imbalance can be ruled out as the cause of the CIC 15-class collapse. The true cause remains an open question rather than an assumed one.

Even where QSVM performs best (EMBER, 15-class, $n_{\text{components}} = 6$), it still trails SVM by 0.27 macro-F1 — the largest classical-quantum performance gap observed in the project to date.

3.2 Angle vs. IQP encoding verdict (CIC 15-class)

The angle and IQP encodings were compared on CIC 15-class, extended to $n_{\text{components}} = 8$ for the first time:

$n_{\text{components}}$	Angle	IQP	Delta (IQP – Angle)
1	0.0550 ± 0.0095	0.0559 ± 0.0077	+0.0009
3	0.0585 ± 0.0057	0.0707 ± 0.0176	+0.0122
6	0.0681 ± 0.0147	0.0794 ± 0.0171	+0.0113
8	0.0693 ± 0.0061	0.0740 ± 0.0103	+0.0046

Table 3: CIC-MalMem 15-class macro-F1: angle vs. IQP encoding (chance ≈ 0.067).

IQP leads at every setting, but every margin is smaller than the fold-to-fold standard deviation of either encoding — a real but weak edge on this dataset. On EMBER, by contrast, the IQP advantage at $n_{\text{components}} = 3$ and 6 is large and clean. The encoding advantage is therefore dataset-dependent rather than an intrinsic property of IQP itself. Increasing to $n_{\text{components}} = 8$ did not rescue CIC 15-class performance (0.0739, barely above the ~ 0.067 random baseline), indicating that qubit count is not the missing ingredient for that collapse.

4 Methodology Notes

- The classical comparison here is SVM only; a broader four-model classical batch (RF/XGBoost/LightGBM in addition to SVM) used in earlier work is out of scope for this branch and should not be compared directly against these numbers.
- Each encoding is run as its own independent invocation, producing a genuinely separate held-out score per encoding rather than a single shared run.
- Quantum kernels are computed first, and classical models replay the same persisted rows/folds via a fixed split-loading mechanism. This replay was confirmed deterministic: reruns reproduced previously committed SVM values to full float precision.
- Inner-cross-validation win-counting (tallying which encoding the tuner selected) has been retired in favor of reporting held-out macro-F1 directly, since win-counting reflects tuner preference rather than the actual magnitude of improvement.

5 Infrastructure and Data Handling

- The published EMBER parquet file is a single row group of $200,000 \times 2381$ float32 values (~ 1.9 GB resident). Column-batched reads keep peak resident memory to ~ 1.14 GiB; without this optimization the sweeps do not fit in available memory.
- A sizing utility solves the measured kernel scaling law for a given runtime budget, recommending $n = 900$ samples at 60s/Gram-matrix and $n_{\text{components}} = 2$. A class-balance floor ($n \geq 750$) for the 15-class task only becomes binding below approximately 40 seconds of budget.
- A SOREL-20M labelling scheme was implemented: dominant tag selected by argmax over raw tag counts prior to binarization, all-zero-tag rows dropped, ties broken by declared column order. Of 19.7M raw rows, 9.57M were successfully labelled across 11 classes; 51.5% were dropped as all-zero-tag, and 13.5% involved tie-breaking. A balanced $n = 900$ subsample can be drawn from the labelled set; a proportionally-weighted subsample cannot, due to a small number of severely over-represented ("flooder") rows.

6 Challenges and Open Issues

- **QSVM underperformance on harder data:** the near-chance QSVM results on EMBER binary classification show that prior ties with classical SVM do not generalize beyond easy, saturated datasets.
- **Unexplained CIC 15-class collapse:** class imbalance and qubit count have both been ruled out as causes, but the actual mechanism behind QSVM's collapse to near-random performance on CIC's 15-class task remains unidentified.
- **SOREL feature acquisition blocked:** the SOREL feature store is a single 71.6 GiB LMDB file with no key-level remote access, meaning no partial subset can be streamed — the entire file must be downloaded locally before any sweep can run. Only the smaller metadata database (3.5 GiB) was fetched, allowing the labelling decision to be closed, but feature acquisition and the associated sweeps remain unstated.
- **CTGAN run blocked:** this remains blocked pending handover of an augmented dataset that does not yet exist in the shared data directory.
- **Environment fragility:** a system-level ROS2 install could leak incompatible pytest plugins into the project's virtual environment, breaking the test runner with a plugin validation error; this was patched with explicit plugin-disabling flags that are no-ops on machines without the conflicting install. The full test suite (119 tests) now passes without workarounds.
- **Duplicated logic risk:** two code paths independently implement the same family-selection rule and can drift out of sync; a docstring fix and a pinning test were added, but the underlying refactor to a single shared function is still deferred.
- **Unverified interactions:** the high all-zero-tag drop rate in SOREL labelling has not been cross-tabulated against the malware/benign flag, and the tie rate has not been sensitivity-checked against tag column ordering.

7 Summary

Binary and multiclass results together show a consistent pattern: QSVM's apparent competitiveness with classical SVM is not intrinsic to the method but an artifact of easy, near-saturated datasets. On a harder binary task (EMBER), and on the hardest multiclass task (CIC 15-class), QSVM either trails SVM substantially or collapses toward chance for reasons not yet explained by imbalance or

qubit count. The IQP encoding shows a real, if dataset-dependent, edge over angle encoding. Progress on SOREL is currently limited to a labelling decision, with feature-level sweeps blocked on a large, non-streamable data file.